Worksheet 4A: Derivatives and Integrals — Inverse Operations

What This Worksheet Is About

Derivatives and integrals are **inverses** of each other, just like multiplication and division, or squaring and square roots.

- **Derivative:** Given a function, find its rate of change
- **Integral:** Given a rate of change, find the original function

But there's a twist: When we "undo" a derivative, we don't get back exactly one function — we get a whole **family** of functions!

Part A: The Power Rule — Both Directions (20 problems)

Differentiation: Bring down the power, reduce by 1

$$\frac{d}{dx}x^n = n \cdot x^{n-1}$$

Integration: Increase the power by 1, divide by new power

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

Notice they're inverses! - Derivative: exponent comes **down** and **decreases** - Integral: exponent goes **up** and we **divide** by it

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Find both the derivative and the integral of each function:

1. $f(x) = x^4$

- $f'(x) =$ _____
- $\int x^4 \, dx =$ _____

2. $g(t) = t^5$

- $g'(t) =$ _____
- $\int t^5 \, dt =$ _____

3. $h(p) = p^3$

- $h'(p) =$ _____
- $\int p^3 \, dp =$ _____

4. $y = x^7$

5. $f(m) = m^2$

6. $g(r) = r^{10}$

7. $h(k) = k^1 = k$

8. $y = x^{100}$

Negative and Fractional Exponents:

9. $f(x) = x^{-2}$

- $f'(x) =$ _____
- $\int x^{-2}\,dx =$ _____

10. $g(t) = t^{-3}$

11. $h(x) = x^{1/2} = \sqrt{x}$

- $h'(x) =$ _____
- $\int x^{1/2}\,dx =$ _____

12. $y = x^{1/3}$

13. $f(p) = p^{-1/2}$

14. $g(n) = n^{3/2}$

The Special Case: What happens when $n = -1$?

15. Try to compute $\int x^{-1}\,dx$ using the formula $\frac{x^{n+1}}{n+1}$.

What goes wrong?

16. The integral of $x^{-1} = \frac{1}{x}$ is actually $\ln|x| + C$. We'll explore why in Worksheet 4D!

Checking Your Work — Derivatives and Integrals Undo Each Other:

17. If $\int x^3\,dx = \frac{x^4}{4} + C$, verify by differentiating: $\frac{d}{dx}\left(\frac{x^4}{4} + C\right) =$ _____

18. If $\frac{d}{dx}(x^5) = 5x^4$, verify by integrating: $\int 5x^4\,dx =$ _____

19. Differentiate, then integrate: Start with $f(x) = x^6$ - $f'(x) =$ _____ - $\int f'(x)\,dx =$ _____ - Do you get back x^6? (Plus what?)

20. Integrate, then differentiate: Start with $g(x) = x^4$ - $\int g(x)\,dx =$ _____ - $\frac{d}{dx}\left[\int g(x)\,dx\right] =$ _____ - Do you get back x^4?

Part B: Why "+C"? — The Family of Antiderivatives (14 problems)

Consider these three functions: - $f(x) = x^2$ - $g(x) = x^2 + 5$ - $h(x) = x^2 - 17$

21. Find the derivative of each: - $f'(x) =$ _____ - $g'(x) =$ _____ - $h'(x) =$ _____

22. What do you notice? Why does this happen?

23. The functions x^2, x^2+5, and x^2-17 are all different, but they have the _____ derivative.

24. Graphically: Sketch $y = x^2$, $y = x^2 + 3$, and $y = x^2 - 2$ on the same axes.

- How are these parabolas related?
- At any given x-value, do they have the same slope?

The Key Insight:

When we integrate $2x$, we get $x^2 + C$ where C is **any constant**.

This represents a whole **family** of functions — infinitely many parabolas, all with the same shape but shifted up or down.

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25. Write three different functions whose derivative is $3x^2$.

26. Write three different functions whose derivative is $5t^4$.

27. Write the general antiderivative (with +C) of $4p^3$.

28. If $F'(x) = 6x^5$, what is $F(x)$?

Finding a Specific Function — Initial Conditions:

If we know $F'(x) = 2x$ AND $F(0) = 7$, we can find the specific function:

$F(x) = x^2 + C$

$F(0) = 0^2 + C = 7$, so $C = 7$

Therefore $F(x) = x^2 + 7$

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29. Find $F(x)$ if $F'(x) = 3x^2$ and $F(0) = 5$.

30. Find $G(t)$ if $G'(t) = 4t^3$ and $G(1) = 10$.

31. Find $H(p)$ if $H'(p) = 2p + 1$ and $H(2) = 0$.

32. A ball's velocity is $v(t) = 20 - 10t$ m/s. Its position is $s(t) = \int v(t)\,dt$.

- (a) Find $s(t)$ with the constant C.
- (b) If the ball starts at height $s(0) = 5$ meters, find C.
- (c) Write the complete position function.

33. An object's acceleration is $a(t) = 6$ m/s². Starting from rest ($v(0) = 0$):

- (a) Find velocity: $v(t) = \int a(t)\,dt$
- (b) If position $s(0) = 10$, find $s(t) = \int v(t)\,dt$.

34. Explain in your own words why integration has "+C" but differentiation doesn't.

Part C: Constants and Sums — Both Directions (16 problems)

Derivative Rules:

- $\frac{d}{dx}(c) = 0$ (constant rule)
- $\frac{d}{dx}[cf(x)] = c \cdot f'(x)$ (constant multiple)
- $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$ (sum rule)

Integral Rules (the reverses):

- $\int 0\,dx = C$ (integrating zero gives a constant)
- $\int c \cdot f(x)\,dx = c \cdot \int f(x)\,dx$ (constant multiple)
- $\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$ (sum rule)

Also: $\int c\,dx = cx + C$ (integrating a constant)

Find both the derivative and the integral:

35. $f(x) = 5x^3$

- $f'(x) =$ _____
- $\int 5x^3\,dx =$ _____

36. $g(t) = -2t^4$

37. $h(p) = \frac{1}{2}p^6$

38. $y = 7$ (a constant function)

- $\frac{dy}{dx} =$ _____
- $\int 7\,dx =$ _____

39. $f(m) = -3$

Sums and Differences:

40. $f(x) = x^3 + x^2$

- $f'(x) =$ _____
- $\int (x^3 + x^2)\,dx =$ _____

41. $g(t) = t^4 - t^2$

42. $h(r) = 3r^2 + 2r - 5$

43. $y = 4x^3 - 6x^2 + 2x - 1$

44. $f(k) = k^5 - 3k^3 + 2k$

45. $g(n) = 2n^4 + n^3 - 5n^2 + 4n - 7$

Verify by reversing:

46. Check problem 40: Differentiate your integral. Do you get back $x^3 + x^2$?

47. Check problem 43: Integrate your derivative. Do you get back $4x^3 - 6x^2 + 2x - 1$ (plus C)?

Rewriting Before Integrating:

48. Find $\int \frac{3}{x^2}\,dx$

Hint: Rewrite as $3x^{-2}$ first

49. Find $\int \frac{5}{\sqrt{t}}\,dt$

50. Find $\int \left(x + \frac{1}{x^2}\right)\,dx$

Part D: Polynomials — Complete Practice (16 problems)

Find the derivative:

51. $f(x) = 2x^5 - 3x^4 + x^3 - 7x + 2$

52. $g(t) = \frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2}$

53. $h(p) = (p + 2)(p - 3)$ *(expand first)*

54. $y = (x - 1)^2$

Find the integral:

55. $\int (3x^2 - 4x + 5)\,dx$

56. $\int (t^4 - 2t^3 + t - 3)\,dt$

57. $\int (6m^5 - 4m^3 + 2m)\,dm$

58. $\int (u^2 + 1)^2\,du$ *(expand first!)*

Mixed — Derivative or Integral as Specified:

59. $\frac{d}{dx}(x^4 - 3x^2 + 2)$

60. $\int (4r^3 - 6r)\,dr$

61. $\frac{d}{dt}(t^6 + 2t^4 - t)$

62. $\int (5k^4 - 3k^2 + 1)\,dk$

Round-Trip Problems:

63. Start with $f(x) = x^5$. - (a) Differentiate: $f'(x) =$ - (b) Integrate your answer: $\int f'(x)\,dx =$ - (c) Is this $f(x)$? Explain.

64. Start with $g(t) = t^3 - 4t$. - (a) Integrate: $\int g(t)\,dt =$ - (b) Differentiate your answer. - (c) Is this $g(t)$? Explain.

65. Why does integrating then differentiating give you back exactly what you started with, but differentiating then integrating gives you the original plus a constant?

66. Complete this analogy:

Squaring and square roots are inverses: $(\sqrt{x})^2 = x$ and $\sqrt{x^2} = |x|$ (almost x, but not quite for negatives)

Similarly: $\frac{d}{dx}\left[\int f(x)\,dx\right] =$ _____ and $\int \frac{d}{dx}[f(x)]\,dx =$ _____

Part E: Building Intuition — What's Really Happening (10 problems)

67. The derivative of position is velocity. The integral of velocity is _____.

68. The derivative of velocity is acceleration. The integral of acceleration is _____.

69. If you know an object's acceleration at all times, can you determine its exact position? What additional information do you need?

70. Match the physical quantities:

Function	Derivative	Integral
Position		
Velocity		
Acceleration		

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71. A car's velocity is $v(t) = 3t^2$ m/s.

- (a) What is the acceleration $a(t) = v'(t)$?
- (b) What is the position $s(t) = \int v(t)\,dt$?
- (c) If $s(0) = 0$, what is the specific position function?
- (d) How far has the car traveled from $t = 0$ to $t = 2$?

72. An object's position is $s(t) = t^3 - 6t^2 + 9t$.

- (a) Find velocity $v(t) = s'(t)$.
- (b) Find acceleration $a(t) = v'(t)$.
- (c) When is velocity zero?
- (d) When is acceleration zero?

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73. Revenue $R(q)$ depends on quantity sold q.

- **Marginal revenue** is $R'(q)$ — the extra revenue from one more unit.
- If you know marginal revenue, $\int R'(q)\,dq$ gives you _____.

74. Cost $C(q)$ depends on quantity produced.

- **Marginal cost** is $C'(q)$.
- If marginal cost is $MC(q) = 6q + 10$, find the cost function $C(q)$.
- If fixed costs are $C(0) = 500$, find the specific cost function.

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75. Population growth rate is $P'(t) = 1000 + 50t$ people per year.

- (a) Find $P(t)$, the population function.
- (b) If the population at $t = 0$ is 50,000, find the specific function.
- (c) What is the population after 10 years?

76. Water flows into a tank at rate $r(t) = 100 - 2t$ liters per minute.

- (a) The amount of water $W(t) = \int r(t)\,dt$. Find it.
- (b) If the tank starts empty, $W(0) = 0$. Find the specific function.
- (c) How much water is in the tank after 10 minutes?
- (d) When does the tank stop filling (when $r(t) = 0$)?

Part F: The Fundamental Theorem Preview (6 problems)

The **Fundamental Theorem of Calculus** connects derivatives and integrals precisely:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where $F'(x) = f(x)$.

This says: To find the integral from a to b, find ANY antiderivative F, then compute $F(b) - F(a)$.

(The $+C$ cancels out!)

Evaluate these definite integrals:

77. $\int_0^2 x^2\,dx$

- Antiderivative: $F(x) = \frac{x^3}{3}$
- $F(2) - F(0) = \frac{8}{3} - 0 =$ _____

78. $\int_1^3 2x\,dx$

79. $\int_0^4 (3t^2 - 2t)\,dt$

80. $\int_{-1}^1 x^3\,dx$

Hint: What symmetry do you notice?

81. $\int_1^4 \sqrt{x}\, dx$

82. $\int_0^3 (x^2 + 1)\, dx$
